Indian Institute of Technology Jodhpur

PG Course Booklet

Indian Institute of Technology Jodhpur

July 2019

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Courses

Offered by

Humanities and Social Science Department

Indian Institute of Technology Jodhpur

PG Compulsory Courses(to be approved)

1. HSN7XX0 Technical Communication 1-0-0 (1)
2. HSN7XX0 Ethics and Professional Life 1-0-0 (1)

PG Electives (to be approved)

1. HSL7XX0 Cine-Politics under National Emergency 3-0-0 (to be approved)
2. HSL7XX0 Sociology of Science and Technology 3-0-0 (to be approved)
3. HSL7XX0 Digital Society 3-0-0 (to be approved)

PG Electives (already approved)

4. HSL7XX0 Gender and Society 3-0-0 (3)
5. HSL7XX0 Sociocultural Theories 3-0-0 (3)
6. HSL7XX0 Foundations of Sustainability 3-0-0 (3)
7. HSL7XX0 Nature Writings in American Literature 3-0-0 (3)
8. HSL7XX0 Literature and Environment 3-0-0(3)
9. HSL7XX0 Twentieth Century Analytic Philosophy 3-0-0 (3)
10. HSL7XX0 Language and Cognition 3-0-0 (3)
11. HSL7XX0 Ethical Theories and Moral Practices 3-0-0 (3)
12. HSL7XX0 Philosophy of Mind 3-0-0 (3)
13. HSL7XX0 Bioengineering Ethics 3-0-0 (3)
14. HSL7XX0 Cognitive Science 3-0-0 (3)
15. HSL7XX0 Qualitative and Quantitative Methodology 3-0-0(3)
16. HSL7XX0 Consciousness 3-0-0 (3)

Title	**Technical Communication**	Number	HSNXX
Department	Humanities and Social Sciences	L-T-P [C]	1–0–0 [1]
Offered for	M.Tech and PhD	Type	Compulsory
Prerequisite	None		

Objectives

The Instructor will:

1. Enable a student to face the challenges in communication, primarily in a technical milieu.

2.Enhance a student's English Language skills namely listening, speaking, reading and writing (LSRW) and the role played by them in the process of communication.

3. Make students conversant in addressing communication challenges that impact organizations through sharing of best practices and group learning.

Learning Outcomes

The students are expected to have the ability to:

1. Comprehend, recognize and articulate the formal elements of specific genres of technical communication.

2.To gain a familiarity with practical and theoretical aspects of various trends and debates in technical communication

3. To develop a sensibility and appreciation for some of the ideas, issues, and problems involved in writing about technology and in workplace writing.

Contents Part I (HSS Faculty) Total – 07 hours (07 Lectures)

Introduction to Communication – Introduction to Communication and the Importance of Communicating in English, Role of Emotions in Communication, Basics of Communication, Purpose and Audience, Cross-Cultural Communication, Language as a tool of communication, LSRW. (1 hour)

Listening -Introduction to Listening, Types of Listening, Active Listening, Barriers to Listening, Listening to pre-recorded audio visual material/film. (1 hour)

Speaking –Introduction to Speaking, Mock Personal Interviews, Body Language, Organizing and learning to make a presentation, Phonetics and Basic Pronunciation, Common Errors in English, Telephone Etiquette, Group Discussions.(2 hour)

Reading - Purpose of Reading, Topic Sentence and its role, Reading for different purposes, Reading Comprehension, Reading Techniques. (1 hour)

Writing– Introduction, Importance, Characteristics, Impersonal and Formal language, Elements of Style, Techniques for Good Technical Writing, Paragraph Writing, Writing a Formal Letter, Writing an Email, Plagiarism. (2 hours)

Part II (Faculty of Engineering and Sciences) – 07 hours (07 Lectures)

Introduction to Technical Communication

Writing – Mechanics of Technical Writing, Organization and Structure in Technical Writing, Writing an abstract, Writing a Research Paper, Writing a Review, Report Writing, Writing Technical descriptions, Copyediting and formatting, Referencing, Citations, Plagiarism.(4 hours)

Reading – Identifying Language Errors, Reading Comprehension with specific reference to passages/articles in Science and Technology. (1 hour)

Lectures with Hands on Training – Grammarly, End note, Language software, Document processing using software.(2 hours)

Reference Books

1.Raman, Meenakshi and Sangeeta Sharma. Technical Communication: Principles and Practice. Oxford University Press: New Delhi, 2011.

2. A Text Book of English for Engineers and Technologists. Division of Humanities and Social Sciences, Anna University, Orient Black Swan: Chennai, 2002.

Online Course Material

1.Technical English for Engineers-https://nptel.ac.in/courses/109106094/

Title	**Ethics and Professional Life**	Number	HSN7XX0
Department	Humanities and Social Sciences	L-T-P [C]	1–0–0 [1]
Offered for	M.Tech and PhD	Type	Compulsory
Prerequisite	None		

Objectives

1. To help students identifying ethical responsibilities linked to their professions
2. To assist students in analyzing professional decisions from ethical perspectives
3. To help students cultivating professional integrity and moral ideals

Learning Outcomes

1. Students will be familiar with the manifold responsibilities attached to their professions
2. Students shall be able toappreciate the role of normative principles in taking professional decisions
3. Students shall be able to respond to ethical issues with integrity and creativeimagination

Contents:

1. *Conceptual Foundations*: Why Should One be Moral?; Relevance of Professional Ethics; Tools and Techniques; Professional Codes; Normativity of Science and Technology (2 hours)

2. *Professions and Moral Dilemmas*: Contemporary Ethical Issues (*to be analyzed with the help of case studies and focusing on the nature of the group*):

Personal: Normative Numbness; Conflicts of Interests; Theft; Deception; Corruption; Partiality

Interpersonal: Contracts; Violation of Rights; Consent and Dissent; Special Obligations; privacy and confidentiality

Organizational: Fairness within the Organization; Hiring; Conduct at Workplace

Social: Norms for a Game Changer; IPR; Reliability of Research Findings; Risk Assessment; Animal and Human Experimentation (09 hours)

3. *Professional Responsibility*: Individual and Collective Responsibility; Corporate Social Responsibility; Rectificatory Issues; Sustainability (3 hours)

Text Books:

1. *Ethical Issues in Professional Life*, Callahan, J., Oxford University Press, 1998.

2. *Philosophy of Technology and Engineering Sciences*, Meijers, A., Elsevier, 2009

Title	Cine-Politics under the National Emergency	Number	**HSL7XX0**
Department	Humanities and Social Sciences	L-T-P [C]	3-0-0
Offered for	B. Tech., M. Tech. and Ph.D. Programs	Type	Elective
Prerequisite	None		

Objectives

1.To help students explore the intricacies of cultural forms and practices in the larger context of the political lives of the Indian nation

2.To help students become familiar with some of the most significant events in the history of post-independence India through the prism of film history and state policy

3.To make students familiar with conceptual apparatuses in cinema studies and with the overlaps between political and cultural histories

Learning Outcomes

The students are expected to have the ability to:

1.Read archival and other materials that are relevant for history-writing

2.Explore associations between public institutions, national history, mass and avant-garde culture and the ever-changing socio-politico-cultural dynamics

3.Develop acquaintance with contemporary scholarly writings on Indian history, cinema and politics

Contents

Introducing Emergency and its Politics–The National Emergency, historical trajectory, relevance in Indian politics, literature review (6 lectures)

Cinema, Cine-Politics and Cultural Politics– Defining Cine-Politics, Nation-Industry negotiations, carrying out cine-archival research, reading of film industry and film society materials/publications, review of relevant theoretical schools in Indian cinema studies (12 lectures)

Sponsorship and Censorship: the Indian New Wave and Other Cinemas- Censorship, Prime Minister's 20-point program, role of I&B ministry, industrial responses, Indian New Wave and state-sponsored cinema in the 1970s, debates on sponsorship and film economy (12 lectures)

Globalization of the Dictatorial in the 1970s: Digital Remembering– The idea of the dictatorial, ideological associations between global dictatorial regimes, cine-cultural evidence and responses, archiving of evidence in the days of the digital, conclusion (12 lectures)

Textbooks

1. Rajadhyaksha, Ashish. *Indian Cinema in the Time of Celluloid: From Bollywood to the Emergency*. Bloomington: Indiana University Press, 2009.

Reference Book

1. Joshi, Priya, and Rajinder Dudrah, eds. *The 1970s and Its Legacies in India's Cinemas*. New York: Routledge, 2014.

Online Course Material

1. https://indiancine.ma/

Title	**Sociology of Science and Technology**	Number	**HSL7XX0**
Department	Humanities and Social Sciences	L-T-P [C]	3-0-0
Offered for	B. Tech, M.Tech and Ph. D.	Type	Elective
Prerequisite			

Objectives

1. To help students conceptualize science and technology as embedded in culture, politics and sociality and understand how technical knowledge is produced in society.

2. To help students explore the dynamic relationship between technology and society and discover how technology and society influence each other.

3. To make students conversant with important and new concepts in science and technology studies in social anthropology.

Learning Outcomes

1.Comprehend technological progress as a social process

2.Find connections between science, technology and culture and think beyond simple binaries like 'natural' and 'social' sciences and 'nature' and 'culture'.

3.Develop sensitivity towards an inclusive, sustainable use of technology and appreciate the significance of diversity in science studies

Contents

Introduction to the Sociology of Science – Introduction to the subject, its founding concepts and traditions, methods of doing STS studies, History of Science and its philosophical traditions – logical positivism, falsification and its critique (5 lectures)

Methodology of Science and its social connections - Structure of scientific revolutions, Ethics and ethos in science and deviance in practice of science, the Matthew Effect, and practicing 'agnostic' science, scientific inventions as cultural capital, social life of scientists. (8 lectures)

Social Construction of Science - Methods of representation in science, Ethnography of a laboratory, social construction of 'realities', social construction of nature, Effect of social understanding of race and gender on production of scientific knowledge (9 lectures)

STS as Social Science **–** Actor Network theory, Pasteurization and the world of non-humans, influence of science studies on social sciences, ecological understanding of the world, towards an inclusive understanding of the universe through STS.(8 lectures)

Technology and New Media – Technology as applied science, Gaming as culture, Hacking as counter-culture, Social media, Algorithms as social documents (8 lectures)

Science and Technology in India- Science & Technology in Colonial India, Development

of Indian Science, Science and Governance in India (4 lectures)

Textbooks

1.Sismondo, S. (2010), *An Introduction to Science and Technology Studies*, 2nd Edition, Wiley Blackwell

2.Miller, D. et. al. edited (2012), *Digital Anthropology,* Berg.

Reference Book

Felt U., Fouche, R. et. al. edited (2017), *The Handbook of Science and Technology Studies*, MIT Press

Online Course Material

1. Mallick, S.*Science, Technology and Society,* NPTEL Course Material, Department of HSS, IIT Guwahati, https://onlinecourses.nptel.ac.in/noc17_hs26/preview

2. Mishra, A.*Sociology of Science,* NPTEL Course Material, Department of HSS, Indian Institute of Technology Roorkee, https://onlinecourses.nptel.ac.in/noc18_hs09/preview

Title	**Digital Society**	Number	HSL7XX0
Department	Humanities and Social Sciences	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Sc., M.Tech, Ph.D.	Type	Elective
Prerequisite	None		

Objectives

The Instructor will:

Introduce students to the dominant discourses that frame debates on digital society

Familiarize students with the literature pertaining to web technologies and their cultural, legal and ethical formations and practices

Familiarize students with the complex relationships between digital cultures and digital divides

Learning Outcomes

The students are expected to have the ability to:

Identify the ways in which digital media shape identity

Utilize new opportunities for meaningful data collection from and using sophisticated forms of artificial intelligence

Identify knowledge and truth amongst the abundance of information

Contents

Introduction to Digital Society: Digital components of a connected society(2 Lectures)

Theorizing Digital Society: New forms of power; Data as sociomaterial objects; Archives; Digital veillance (6 Lectures)

Digital Identities and Relationships: Self and the Digital Society; Embodied Identities in Digital Society; Bias and Privilege – Digital Inequalities; Marginalised Histories**;** Cyborgs(6 Lectures)

Digital Spaces and Practices: Rethinking space and surveillance in digital societies; Gender, Space, and Place in Digital Societies**;** Urban Informatics and Sociological Imagination - Smart cities**;** Digital Healthcare; Mobility in Digital Society; Digital Heritage (10 Lectures)

Network Society: The Internet as a Network; Networks and the Cultural Imaginary; Inequalities in the Network Society; Information Capital; Interface Design for Diverse Populations (10 Lectures)

Re-conceptualizing Research in a Digital Age: Information Management – Data Analysis Software**;** Large Digital Systems; Data protection and the politics of data privacy (8 Lectures)

Textbook

1. Lupton,D., (2015), *Digital Sociology*, London, New York:Routledge

2. Gere,C., (2008), *Digital Culture*, 2nd Edition, London: Reaktion Books Limited

Reference Books

1. Bentkowska-Kafel, A., Cashen, T., and Gardiner, H. (Eds.) (2009), *Digital Visual Culture: Theory and Practice*, Bristol and Chicago: Intellect Books

2. Karaganis, J. (Ed.), (2007), *Structures of Participation in Digital Culture*, Social Science Research Council, Columbia University Press

3. Tredinnick, L. (2008), *Digital Information Culture: The Individual and Society in the Digital Age*, Oxford: Chandos Publishing Limited

Online Resources

1. Digital Humanities Seminar Video Archive of the Open University, UK, http://www.open.ac.uk/arts/research/digital-humanities/videos